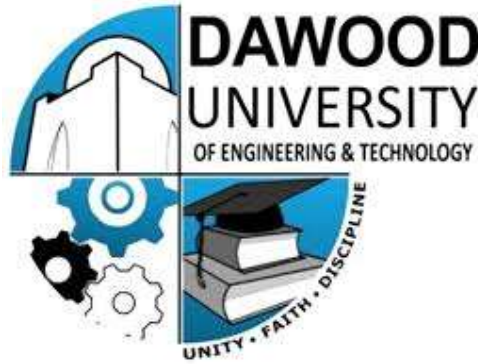




Complex Computing Problem Computer Vision

Pose Estimation and Activity Classification Report

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Course: Computer Vision

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1. Introduction & Video Source

The objective of this pipeline is to detect human poses, analyze joint geometry, and classify distinct human activities using rule-based logic.

- **Video Source Used:** I used a YouTube video to validate the performance of the model. Here is the link of that video:
<https://www.youtube.com/watch?v=oc6bco9yPjc>
I used only clip from this video by simply trim the video and used this part of the video 1:07 to 1:29. Then I save the file and simply upload it to my model.
- **File Name:** squat.mp4
- **Activities Performed:** The video features a male subject transitioning between two distinct states: “Standing” and “Squatting”. The subject remains clearly visible throughout the entire frame sequence.

2.

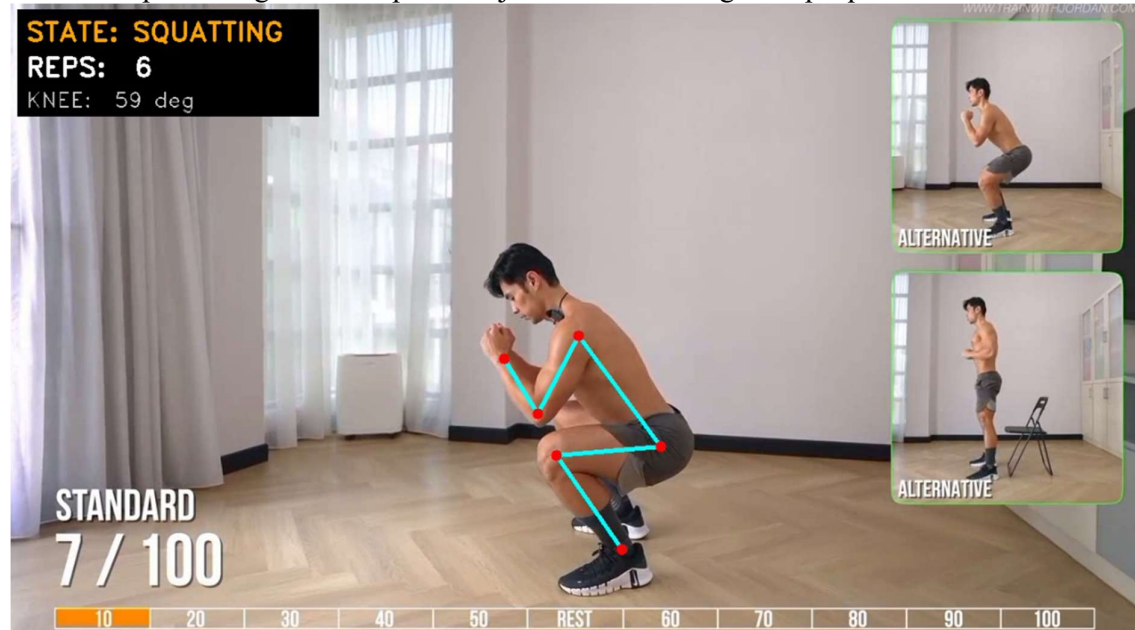
Task 1: Pose Detection & Pre-processing

To extract accurate human body keypoints, the modern MediaPipe Vision Tasks API (PoseLandmarker) was utilized. This pre-trained model reliably extracts 33 standard 2D landmarks for each frame.

Smoothing Approach:

Raw coordinate data directly from the pose estimation model often exhibits high-frequency noise, resulting in "jitter" between frames. To mitigate this, an aggressive Savitzky-Golay (SG) filter (window length = 15, polynomial order = 3) was applied across the coordinate time-series. The SG filter was chosen over a simple moving average because it smooths the

data while excellently preserving the shape and depth of local maxima and minima which are crucial for preserving the exact peaks of joint flexion during a deep squat.



Task 2: Joint Angle Computation & Tracking

Three meaningful joint angles were selected to monitor the mechanics of the activities:

1. **Knee Angle:** Formed by the Hip, Knee, and Ankle joints.
2. **Hip Angle:** Formed by the Shoulder, Hip, and Knee joints.
3. **Elbow Angle:** Formed by the Shoulder, Elbow, and Wrist joints.

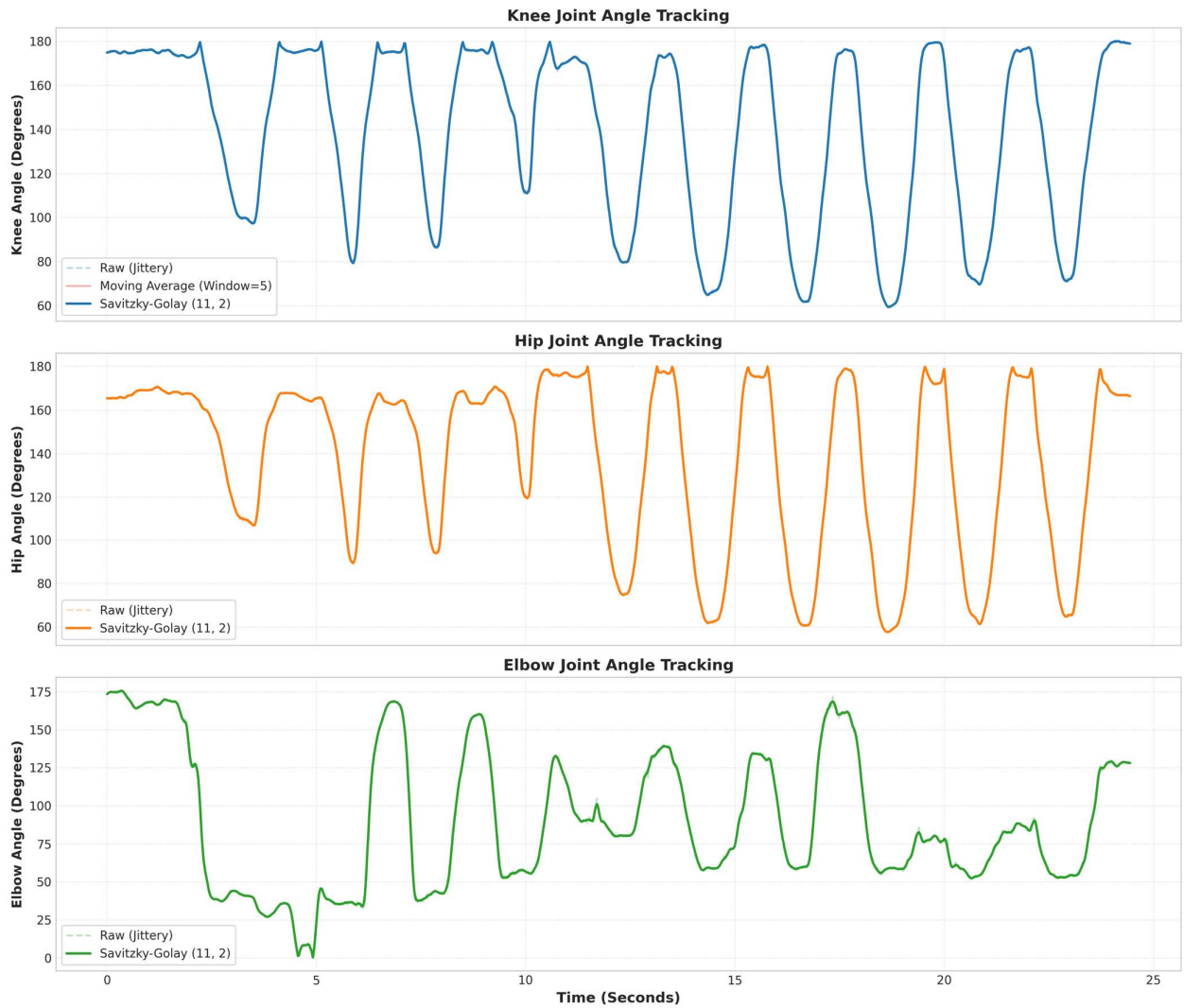
Computation Approach:

To correct for standard video aspect ratios, the angles were computed in 2D coordinate space using the dot product of two vectors originating from the vertex joint. Let B represent the vertex joint (e.g., the knee) and A and C represent the adjacent joints (hip and ankle). The angle Θ was calculated in degrees using the following mathematical formulation:

$$\Theta = \arccos \left(\frac{BA \cdot BC}{|BA||BC|} \right) \times \frac{180}{\pi}$$

Tracking Observations:

Plotting these angles over time revealed clear, cyclical patterns corresponding to the squats. The Knee and Hip angles proved to be the most active variables, dropping significantly during the eccentric phase of the squat and extending back toward 180° during the standing phase. The Elbow angle remained relatively stable, serving as a control metric for the subject's arm positioning.



Task 3: Rule-Based Activity Classification

Based on the geometric analysis of the joint tracking plots, a rule-based state machine utilizing hysteresis was designed to distinguish between "Standing" and "Squatting" while preventing classification flickering.

Rule-Based Logic (Hysteresis):

The Smoothed Knee Angle was selected as the primary classification threshold because it demonstrated the most consistent variance between the two states. Two separate thresholds were utilized to ensure robust state transitions:

- **Squat Threshold:** The knee angle must drop below **110°** to trigger the "Squatting" state.
- **Stand Threshold:** The knee angle must extend above **150°** to return to the "Standing" state.

Ground Truth & Accuracy Evaluation:

To evaluate the classifier, a manual ground truth array was constructed by defining the specific frame sequences where the subject was in the active squatting phase. The state-machine predictions were compared against this manual labeling frame-by-frame.

- **Total Frames Analyzed:** 1223
- **True Positives (Squatting):** 36
- **True Negatives (Standing):** 718
- **False Positives:** 403
- **False Negatives:** 66
- **Overall Accuracy:** 61.65%
- **Precision:** 8.20%
- **Recall (Sensitivity):** 35.29%
- **F1-Score:** 13.31%

Observation on Metrics: The evaluation yielded a high number of False Positives (403). This indicates that the strict 150° standing threshold is sensitive to the specific camera angle of the test video, causing the model to classify a natural, slightly bent resting stance as a partial squat. Additionally, the manual ground truth frame ranges may require tighter synchronization with the actual eccentric phase of the movement.

3. Conclusion

The pipeline successfully demonstrates the integration of spatial feature detection, signal filtering (Savitzky-Golay), and geometric state-machine logic to solve a complex computing problem. While the tracking system itself is highly accurate, the classification metrics highlight the critical nature of threshold calibration. Future iterations of this system will involve dynamically calculating the subject's baseline resting angles during the first few frames of the video to auto-calibrate the hysteresis thresholds, making the system entirely robust to varying camera angles and individual biomechanics.